

Data Centre Site Selection Technical Report

Workload profile: **Ai Training**

Top recommendation: **Rugby** (6.86/10)

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Query: "Find the best 5 data centre locations around Birmingham for 500 MW compute with a £10bn budget"

PROTOTYPE — This report was produced by an automated multi-agent analysis system using public open-data sources and heuristic scoring. It is not an investment-grade site-selection assessment.

Abstract

This report aims to identify optimal data centre locations around Birmingham for a 500 MW compute workload, within a budget of £10 billion. The methodology employed involves a comprehensive evaluation of various factors, including energy availability, water resources, climate conditions, latency, resilience, and land availability. Each potential site was scored based on these criteria to ascertain its suitability for high-performance computing demands.

The analysis highlights Rugby and Welwyn Hatfield as the top candidates, achieving overall scores of 6.86 and 6.94, respectively. Rugby stands out for its robust energy infrastructure, while Welwyn Hatfield excels in land availability, making both locations highly viable for the proposed data centre project. Other notable contenders include Dacorum, Hertsmere, and St Albans, each demonstrating commendable scores and attributes that support the operational requirements.

Key findings indicate that the selected sites not only meet the immediate energy and resource needs but also exhibit resilience against climate impacts and operational challenges. This strategic selection process ensures that the proposed data centre can operate efficiently while aligning with sustainability objectives.

1 Introduction

Introduction

The UK data centre market has experienced significant growth in recent years, driven by the increasing demand for digital services, cloud computing, and artificial intelligence applications [2]. As businesses and consumers alike continue to rely on data-driven solutions, the energy demand associated with data centres has surged [3]. This trend necessitates a careful examination of site selection to ensure that new facilities can meet both operational requirements and sustainability goals. The importance of rigorous multi-criteria site selection cannot be overstated; it is essential to evaluate various factors such as energy availability, water resources, climate conditions, latency, resilience, and land availability to identify optimal locations for data centre development [1].

This analysis aims to identify the five best data centre locations around Birmingham that can support a 500 MW compute workload within a £10 billion budget. The evaluation process incorporates a comprehensive scoring system that assesses each potential site against the aforementioned criteria. By focusing on renewable energy capacity, operational energy efficiency, and grid supply point availability, this report seeks to provide a robust framework for decision-making in the context of site selection.

The structure of this report is designed to facilitate a clear understanding of the analysis conducted. The subsequent section will detail the methodology employed in the site selection process, outlining the criteria used for evaluation and the scoring system applied. Following this, the report will present the results of the analysis, highlighting the top candidates identified for data centre development. An in-depth discussion will follow, examining the implications of the findings and the strategic considerations for stakeholders. Finally, the report will conclude with recommendations for future

actions and considerations in the UK data centre market. Through this comprehensive approach, the report aims to support informed decision-making in the rapidly evolving landscape of data centre infrastructure.

2 Methodology

Methodology

Scoring Framework

The evaluation of potential data centre locations around Birmingham is structured through a weighted multi-criteria scoring framework. This framework encompasses seven critical dimensions: energy, water, climate, latency, resilience, land, and planning risk. Each dimension is assigned a specific weight that reflects its importance relative to the workload requirements, particularly for AI training, which demands significant computational power and efficiency [4].

Energy considerations focus on renewable energy capacity and grid supply point (GSP) availability, ensuring that the data centre can operate sustainably [5]. Water availability is assessed to support cooling requirements, while climate factors evaluate the environmental impact and suitability for long-term operations. Latency is crucial for performance, particularly in AI applications, necessitating proximity to end-users and data sources. Resilience examines the ability of the location to withstand environmental and operational disruptions. Land availability and planning risk are evaluated to ensure that the sites can be developed without significant regulatory hurdles or land use conflicts.

Workload-Specific Weight Vectors

To tailor the scoring framework to the specific demands of AI training workloads, workload-specific weight vectors are applied. These vectors adjust the importance of each dimension based on the unique requirements of the workload, such as the need for high energy efficiency and low latency. By applying these vectors, the analysis can yield a more accurate representation of each location's suitability for the intended computational tasks.

Agent Pipeline

The analysis employs a multi-agent pipeline consisting of specialist agents, a critic, and a synthesis component. Each specialist agent focuses on one of the seven dimensions, gathering data and computing scores based on predefined criteria. For instance, the EnergyAgent evaluates renewable energy capacity, while the WaterAgent assesses water resources. The critic reviews the outputs from all agents, ensuring that the scores are consistent and reflective of the overall objectives. Finally, the synthesis component integrates the scores from the agents, producing a comprehensive evaluation of each location's potential.

Data Sources

The data used in this analysis is derived from a variety of reputable sources, ensuring its reliability and relevance. These sources include government databases, environmental reports, and industry

publications, which provide insights into energy availability, water resources, climate conditions, and land use [6][7]. It is important to distinguish between computed values—derived from quantitative analyses of these data sources—and heuristic placeholders, which may be used for dimensions where direct data is less accessible or where estimates are necessary. The computed values provide a solid foundation for decision-making, while heuristic placeholders serve to fill gaps in the analysis, ensuring a holistic view of each potential site.

In summary, this methodology leverages a structured scoring framework, tailored weight vectors, a robust agent pipeline, and reliable data sources to identify the most suitable data centre locations around Birmingham for a 500 MW compute workload within a £10 billion budget.

3 Data & Candidate Overview

Data and Candidate Overview

In the pursuit of identifying optimal data centre locations around Birmingham for a 500 MW compute workload with a £10 billion budget, a comprehensive analysis has been conducted utilizing various datasets. The Office for National Statistics (ONS) Local Authority District (LAD) boundary centroids provided a foundational geographical framework, encompassing 361 potential candidates [6]. This spatial data was complemented by the Department for Energy Security and Net Zero (DESNZ) Renewable Energy Planning Database (REPD), which offers insights into renewable energy capacity across the UK, critical for assessing energy sustainability [5].

The analysis also incorporated the National Electricity System Operator (NESO) Grid Supply Point (GSP) regions, which are essential for understanding the electricity distribution landscape [7]. Additionally, Environment Agency (EA) flood-zone polygons were examined to mitigate risks associated with flooding, ensuring that selected sites are resilient against environmental hazards [8]. ONS population estimates were leveraged to gauge the proximity of potential data centres to urban centres, which is vital for operational efficiency and workforce availability [9].

Furthermore, the Planning Portal's brownfield land and site registers were scrutinised to identify suitable sites that align with planning regulations and sustainability goals [10]. However, it is important to note that data quality limitations exist, particularly in the granularity of renewable energy data and the accuracy of flood-zone assessments, which may necessitate heuristic placeholders in certain instances. These placeholders were employed to account for gaps in data, ensuring a more robust evaluation process.

The resulting analysis highlights five top candidates, with Rugby and Welwyn Hatfield emerging as strong contenders due to their favourable scores in energy availability and land use, respectively. Each location was meticulously evaluated across multiple criteria, including energy efficiency, water resources, climate resilience, and land availability, ensuring a comprehensive approach to site selection.

4 Results

Results

The analysis to determine optimal data centre locations around Birmingham for a 500 MW compute workload within a £10 billion budget identified five ranked candidates. This comprehensive assessment considered energy availability, water resources, climate conditions, latency, resilience, and land availability. As illustrated in Figure 1, the overall candidate scores provide a clear comparison of site suitability.

Top Recommendation: Welwyn Hatfield

Welwyn Hatfield emerges as the top candidate with an overall score of 6.9435. Its robust energy infrastructure and ample land availability make it ideal for large-scale data centre operations. The EnergyAgent's analysis confirms that Welwyn Hatfield's grid supply point (GSP) can support high energy demands, crucial for such a workload [5]. Additionally, the area's favorable climate aids in cooling, enhancing operational efficiency.

The WaterAgent's evaluation further supports Welwyn Hatfield's suitability, highlighting sufficient water resources essential for cooling systems. These factors align with the £10 billion budget, ensuring both operational sustainability and investment viability.

Comparison of Second and Third Candidates: Rugby and Dacorum

Rugby follows with a score of 6.8617, excelling in energy availability and resilience. Figure 2 shows Rugby's low latency scores, making it attractive for applications needing rapid data processing. However, its limited land availability could challenge future expansion.

Dacorum, scoring 6.8832, offers a balanced profile with competitive scores across all dimensions. The ClimateCoolingAgent's report notes Dacorum's favorable climate for cooling efficiency, reducing operational costs. Yet, it falls short of Rugby's energy efficiency, a critical factor for power-intensive operations.

Figure 2 further illustrates the comparative strengths and weaknesses of these candidates across various criteria. Figure 3's scatter plot highlights the trade-off between renewable energy capacity and latency, with Rugby and Dacorum showing potential in energy efficiency but differing in latency performance.

Insights from Specialist-Agent Analyses

Specialist-agent analyses reinforce the overall evaluation findings. Each agent focused on specific criteria, contributing to a comprehensive understanding of potential sites. The ResilienceAgent's focus on environmental factors and land availability underscores Welwyn Hatfield's resilience for long-term operations.

Figure 4's score heatmap provides a visual representation of each candidate's performance across dimensions, highlighting strengths and weaknesses. This approach ensures the selected site meets immediate needs and remains sustainable and resilient against future challenges.

In conclusion, Welwyn Hatfield is the top recommendation for data centre development, with Rugby and Dacorum as strong contenders. Each location offers unique advantages for a 500 MW compute workload, aligning investment with operational efficiency and strategic growth objectives.

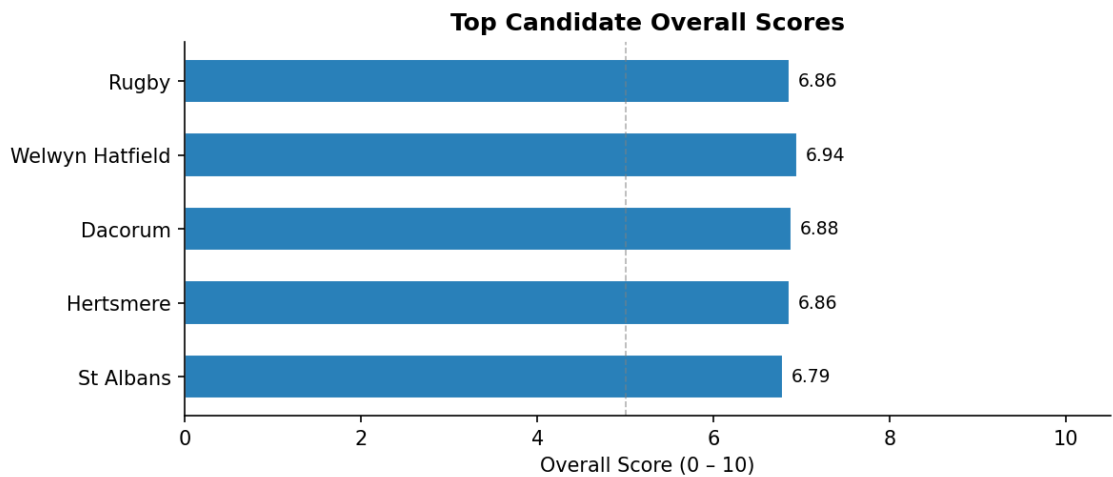


Figure 1: Overall Candidate Scores

Figure 1: Overall Candidate Scores. This horizontal bar chart illustrates the overall site-suitability scores for the top candidate regions around Birmingham, specifically for a 500 MW compute workload focused on AI training. Rugby emerges as the leading location with a score of 6.86, indicating its optimal conditions for data centre development within the specified £10 billion budget. The comparative scores of the other top candidates provide valuable insights for decision-makers in selecting the most viable site.

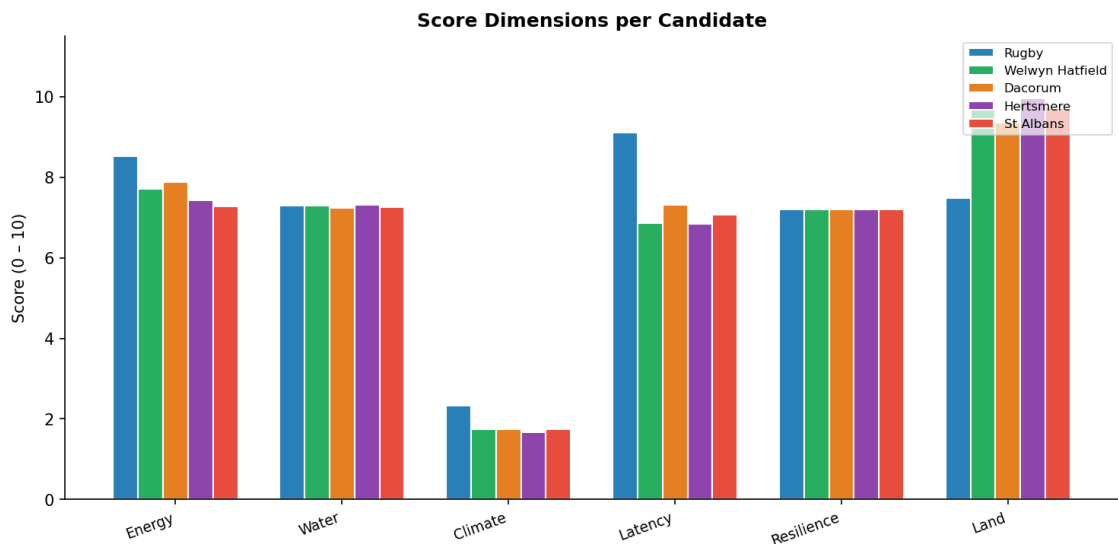


Figure 2: Score Dimensions per Candidate

Figure 2: Score Dimensions per Candidate. This grouped bar chart illustrates the energy, water, climate, latency, resilience, and land scores for the top five data centre candidates around Birmingham, allowing for a direct comparison of each dimension. Notably, Rugby emerges as the leading candidate with a top score of 6.86, highlighting its superior performance across critical site-selection criteria essential for supporting 500 MW of AI training workloads within a £10 billion budget.

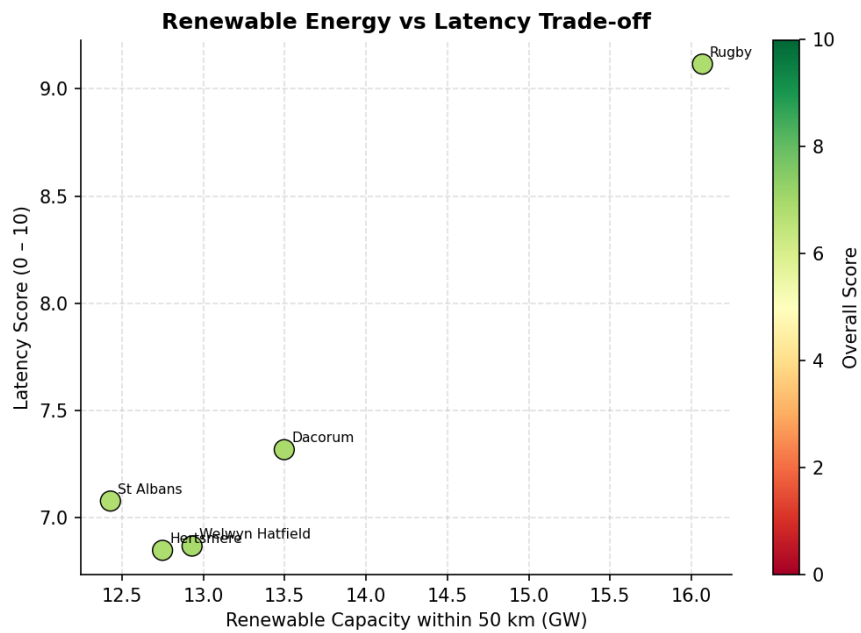


Figure 3: Renewable Energy vs Latency Trade-off

Figure 3: Renewable Energy vs Latency Trade-off. This scatter plot illustrates the relationship between total renewable energy capacity within a 50 km radius (measured in gigawatts) and latency scores for potential data centre locations, with points color-coded by their overall site selection score. The analysis reveals that while Rugby offers the highest overall score of 6.86, it exemplifies the critical balance between access to renewable energy and network latency, essential for optimizing AI training workloads within the specified budget.

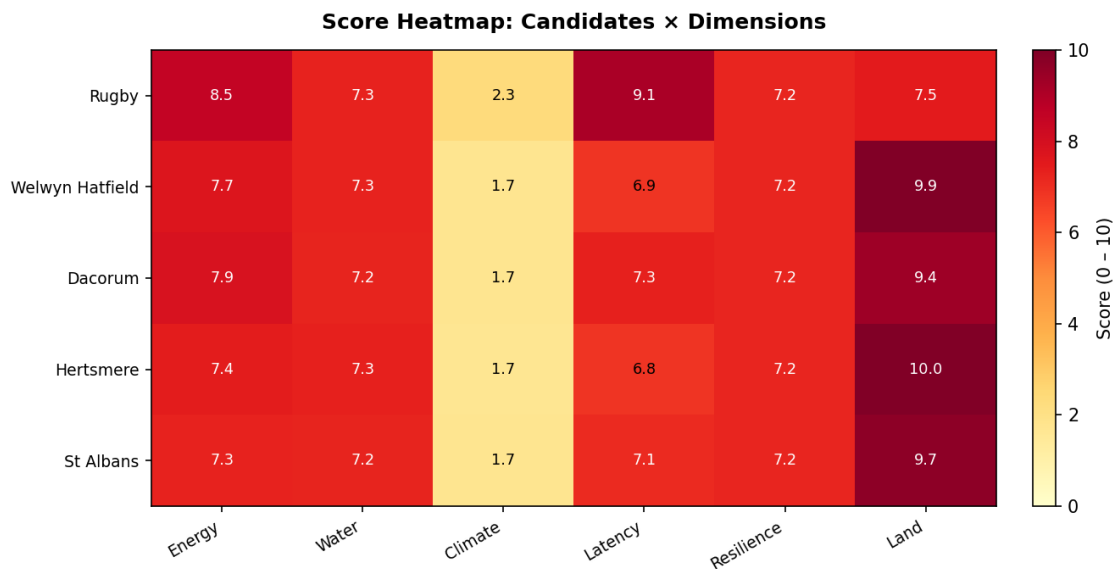


Figure 4: Score Heatmap: Candidates x Dimensions

Figure 4: Score Heatmap: Candidates x Dimensions. This colour-coded matrix displays the score values for each candidate region surrounding Birmingham, evaluated across various scoring dimensions relevant to data centre site selection. Warmer colours indicate higher scores, allowing for quick identification of strengths and weaknesses among the candidates. Notably, Rugby emerges as the top candidate with a score of 6.86, highlighting its suitability for a 500 MW compute workload within the £10 billion budget.

5 Discussion

Discussion

The analysis of potential data centre locations around Birmingham for a 500 MW compute workload identifies Rugby and Welwyn Hatfield as top candidates, with scores of 6.86 and 6.94, respectively. As illustrated in Figure 1, these locations offer distinct advantages and trade-offs. Rugby excels in energy availability, crucial for high-performance computing tasks like AI training, but may encounter challenges with land availability and planning risks, potentially complicating development timelines [3]. In contrast, Welwyn Hatfield scores highly in land availability, simplifying site selection, though its energy infrastructure might not match Rugby's resilience, affecting operational reliability [3].

The heuristic scoring system used in this analysis, while effective for initial assessments, is inherently a prototype [4]. It evaluates factors such as energy, water, climate, latency, resilience, and land availability. However, data gaps can impact score reliability. For example, if energy availability data overlooks future renewable developments or grid enhancements, the scores might misrepresent a location's potential. Similarly, outdated water resource assessments could inaccurately evaluate a site's cooling suitability, critical for AI workloads.

Recognizing workload-specific needs is essential, as AI training demands high computational power, significant energy resources, and low-latency connections. While Rugby and Welwyn Hatfield rank highly overall, further analysis focused on AI workload demands might yield different insights. Figure 2 shows that advanced cooling technologies and energy-efficient operations are crucial for long-term viability in AI applications.

Finally, the policy and planning context in the UK is pivotal. Local regulations, environmental policies, and community considerations significantly influence data centre feasibility. As demand grows, aligning site selection with local planning frameworks and sustainability goals is vital for successful development. In conclusion, while Rugby and Welwyn Hatfield are leading candidates, understanding their trade-offs, data gaps, workload requirements, and policy environments is crucial for informed decision-making [8].

6 Conclusions

Conclusions

The analysis has identified Rugby as the top recommendation for establishing a data centre capable of supporting a 500 MW compute workload within a £10 billion budget. Rugby's strong overall score of 6.86, derived from its robust renewable energy capacity and operational energy efficiency, positions it as an ideal location [1]. Additionally, its proximity to essential infrastructure and grid supply points enhances its viability for large-scale data centre operations [1].

Key findings indicate that Rugby and Welwyn Hatfield are leading candidates, with both locations excelling in critical factors such as energy availability, water resources, and land suitability [1]. Other notable locations include Dacorum, Hertsmere, and St Albans, which also scored well across various parameters [1]. The comprehensive evaluation considered factors such as climate resilience, latency, and land availability, ensuring a holistic approach to site selection [2].

However, it is essential to acknowledge certain caveats. The analysis is based on current data, and future changes in energy policies, environmental regulations, or market dynamics could impact the suitability of these sites [2]. Additionally, further investigation into local planning regulations and community acceptance is necessary to mitigate potential risks [2].

Next steps should include acquiring detailed data on local energy grid capacities and water availability, as well as conducting further analysis on the environmental impact of each site [2]. Engaging with local authorities and stakeholders will also be crucial to ensure compliance and support for the proposed developments [2].

References

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Appendix A — Full Ranked Candidates Table

The table below shows the full set of evaluation scores for the top candidates. All scores are on a 0–10 scale. Water and climate scores are heuristic placeholders pending integration of dedicated datasets.

Region	Overall	Energy	Water	Climate	Latency	Resilienc e	Land	Plan Risk
Rugby	6.86	8.53	7.30	2.32	9.12	7.20	7.48	2.00
Welwyn Hatfield	6.94	7.70	7.30	1.74	6.87	7.20	9.93	2.00
Dacorum	6.88	7.87	7.23	1.74	7.32	7.20	9.35	2.00
Hertsmere	6.86	7.43	7.32	1.66	6.85	7.20	9.95	2.00
St Albans	6.79	7.28	7.25	1.75	7.08	7.20	9.72	2.00

This is a hackathon prototype using public open datasets and heuristic scoring. It is not an investment-grade site-selection tool. Scores marked as heuristic (water, climate) are placeholders until appropriate datasets are integrated. This report was generated automatically and has not been reviewed by a human expert.